

Structured Electronics Design

Review | Group 2

18-5-2026



Outline

1) Specification/
concept design

2) Feedback
configurations

3) First stage/
noise design

4) Single stage

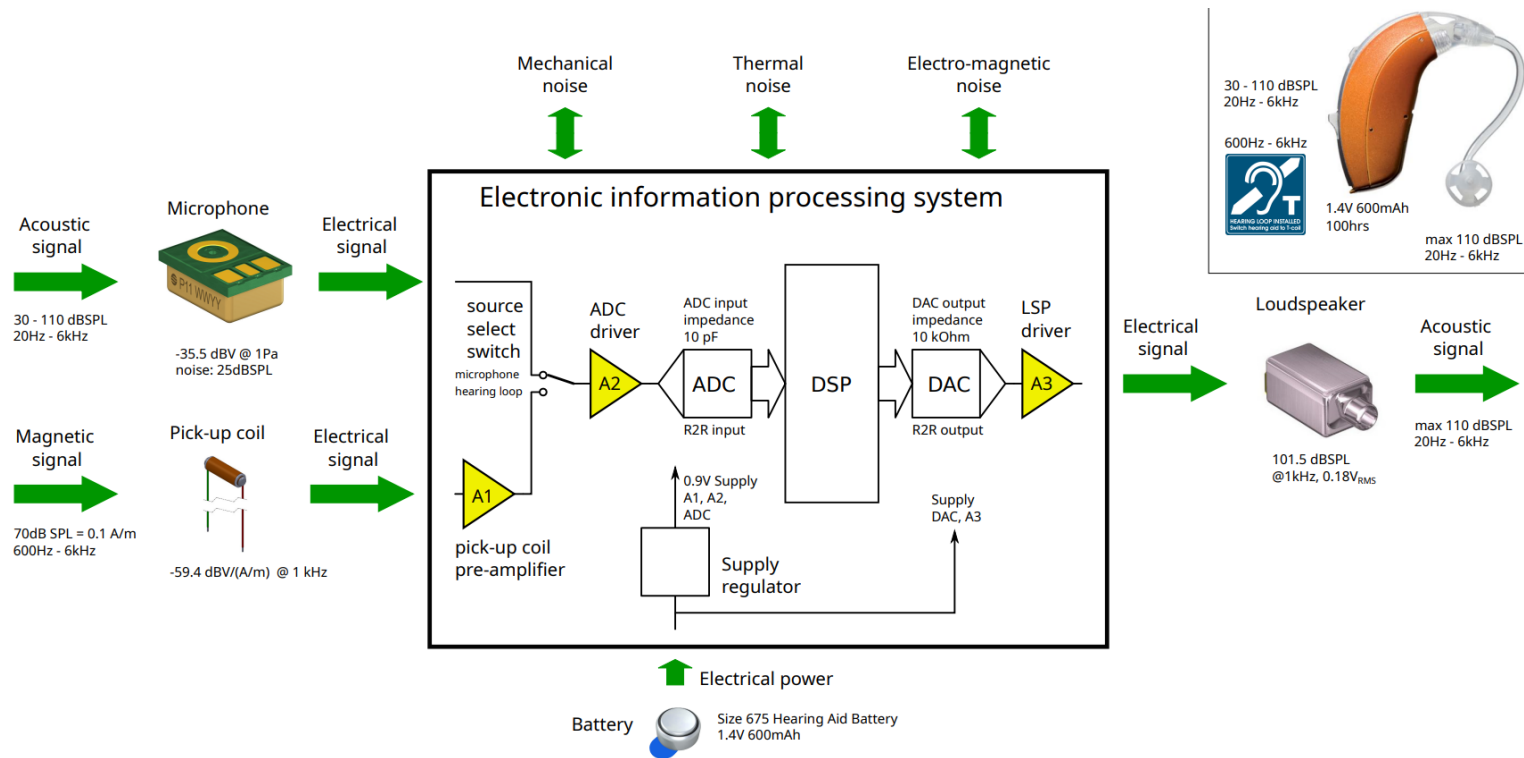
5) Dual stage

6) Frequency
Compensation

Specification

What should we do before to construct the specification?

1. In what environment our circuit work? What components our circuit is interacting with?
2. Understand how does our circuit work
3. Set the specification and budget



Input:

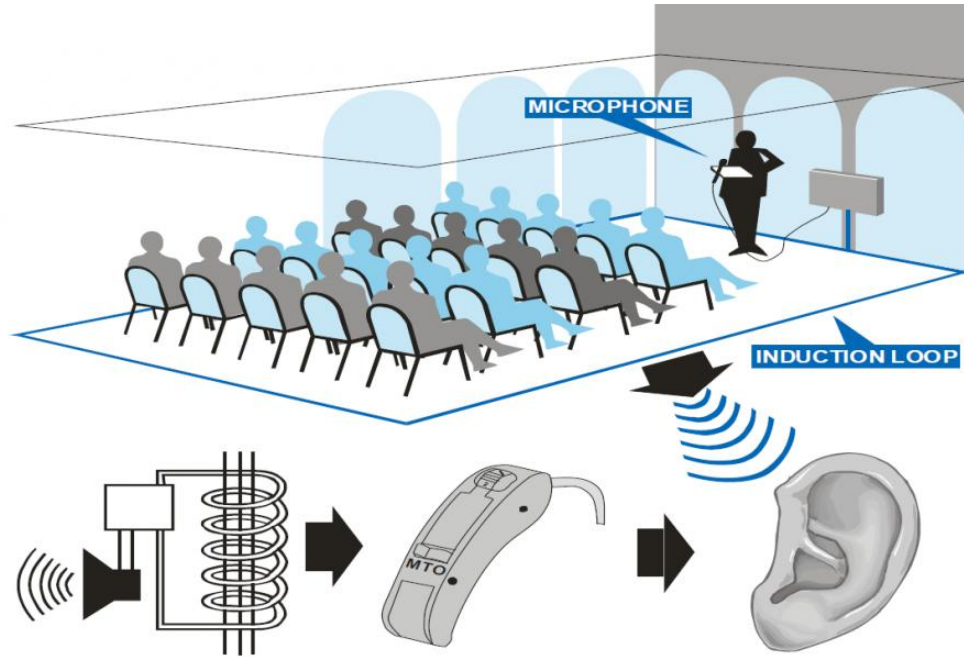
Pick up coil

Output:

1. Source select switch (Microphone/hearing loop)
2. ADC with input impedance of 10pF

Input of A1: Pick up coil

How does it work? Differentiator



<https://www.majorcom.fr/en/induction-loop-systems/>

Current passes through Induction Loop

Magnetic field is generated around it

Pick up coil is brought within this magnetic field

A corresponding current is created

Voltage is generated proportional with the rate of change of magnetic flux over the time

$$V = -\frac{d\phi}{dt}$$

Output of A1

Source select switch (Microphone/hearing loop)

- Allow switching between the output of microphone and hearing loop (pick up coil)
- Requires **$V_{\text{microphone}} = V_{\text{hearing loop}}$ @reference frequency**

ADC with input impedance = 10pF

- ADC (A2) works as voltage amplifier input
 - $Z_i = \infty$
 - draws no current from A1's output
- The only loading effect left is the parasitic capacitance of the ADC input (10pF)

How would A1 work?

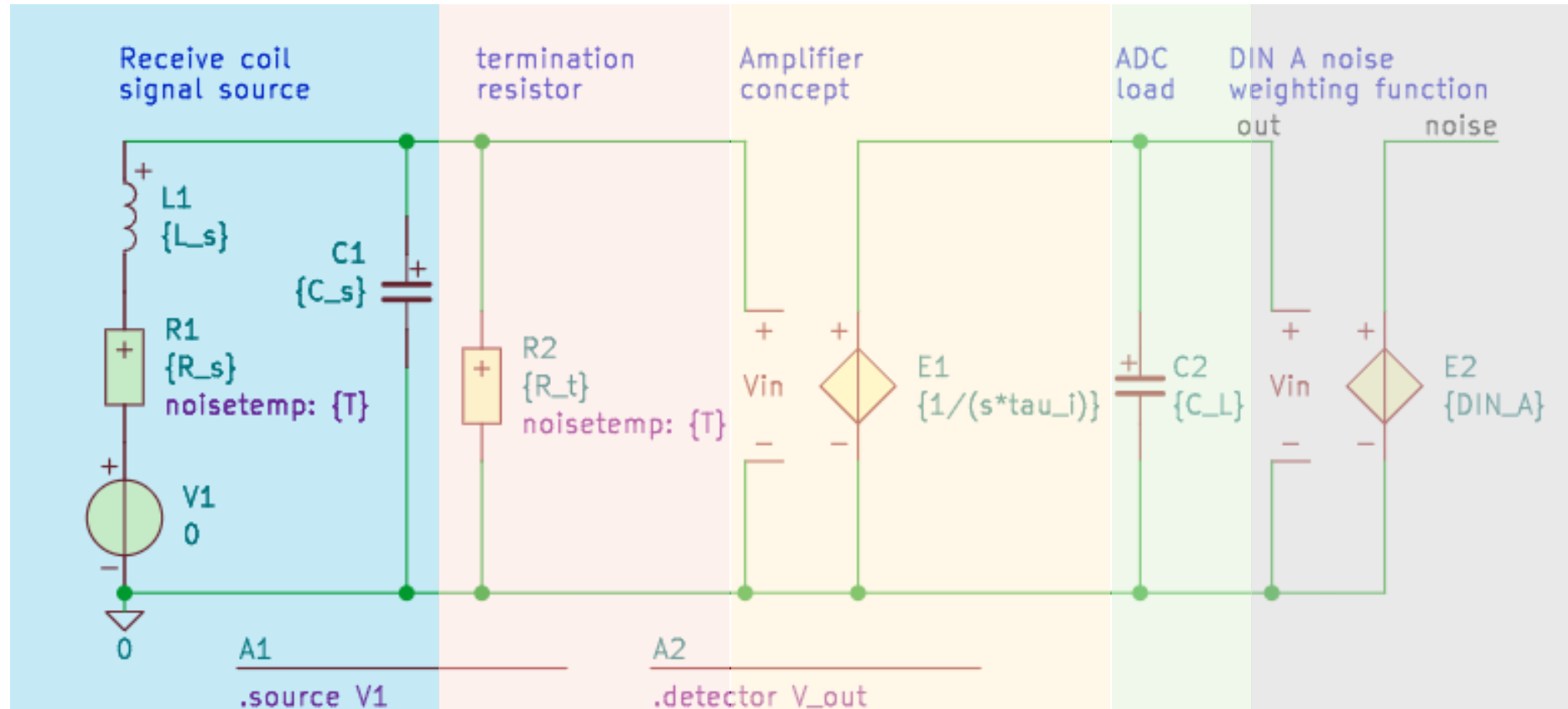
1) A1 will work as **an integrator**

2) A1 is a **voltage-voltage amplifier**

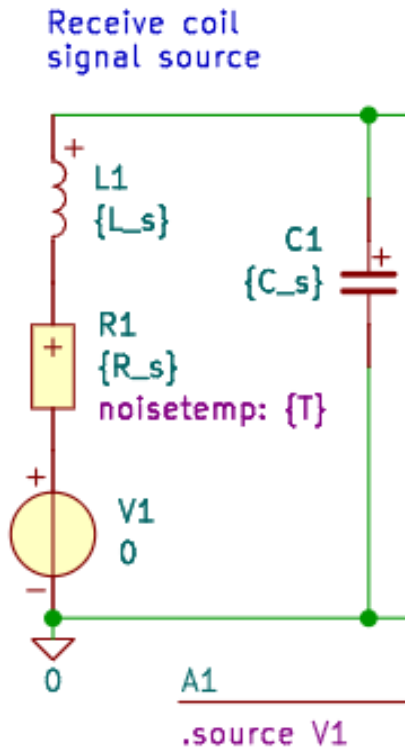
3) A1 has a **specific gain magnitude**, applies when $V_{\text{microphone}} = V_{\text{pick up coil @reference frequency}}$

4) Noise requirement, where **Vnoise (after DIN-A weighting) does not exceed the microphone's noise floor**

Circuit Model



Pick Up Coil Representation



Electrical data

Measuring conditions: Temperature 23°C (73.4F) ± 5°C (64.4F-82.4F)

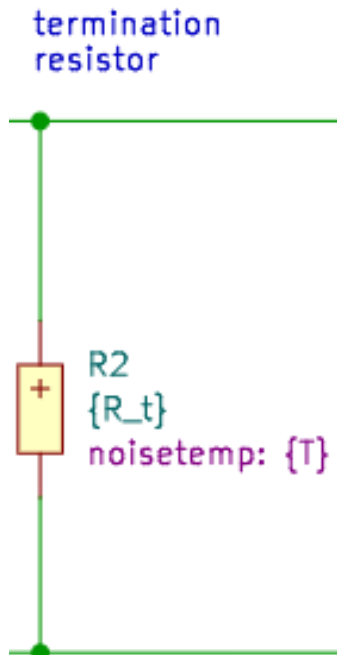
Parameters	Min	Typ	Max	Unit	Comments
DC resistance	788	875	962	Ohm	
Inductance @ 1 kHz	102	120	138	mH	
O.C. 1 kHz sensitivity	-60.9	-59.4	-57.9	dBV/A/m	
10 kOhm load 1 kHz sensitivity	-61.7	-60.2	-58.7	dBV/A/m	
Resonance frequency	150			kHz	
Winding sense	North positive				

$$f_{res} = \frac{1}{2\pi\sqrt{L_s C_s}}$$

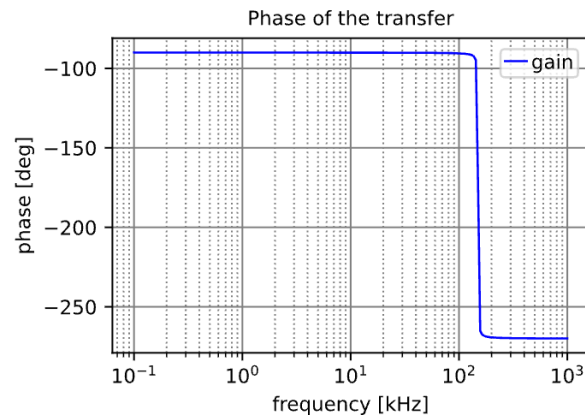
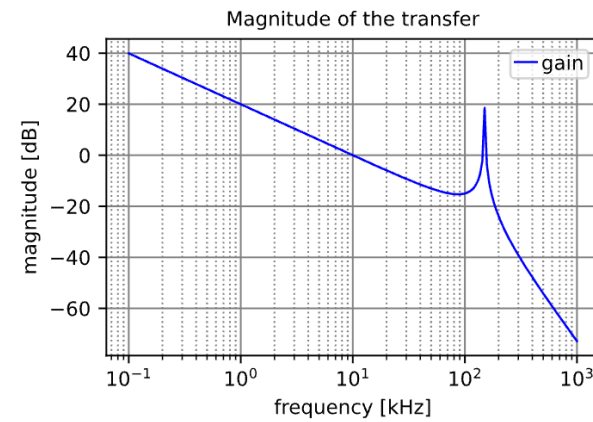
$$C_s = \frac{1}{(2\pi f_{res})^2 L_s} = \frac{1}{(2\pi \cdot 150 \text{ kHz})^2 \cdot 120 \text{ mH}} = 9.382 \cdot 10^{-12} \text{ F}$$

Termination Resistor Representation

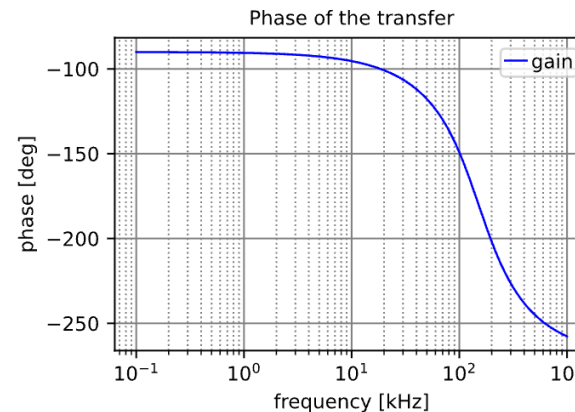
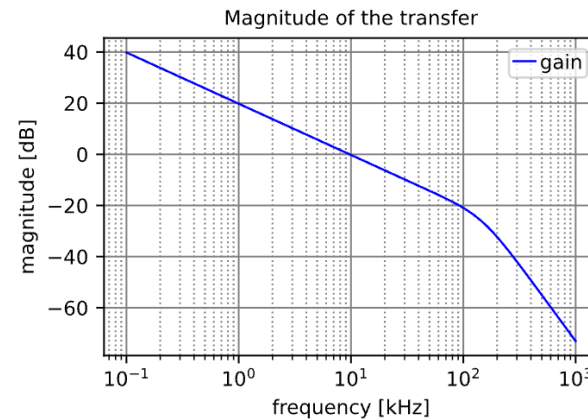
Purpose: Damp LC resonance (properties of pick up coil)



Without termination resistor



With termination resistor



In critically damping condition (R_t optimal)

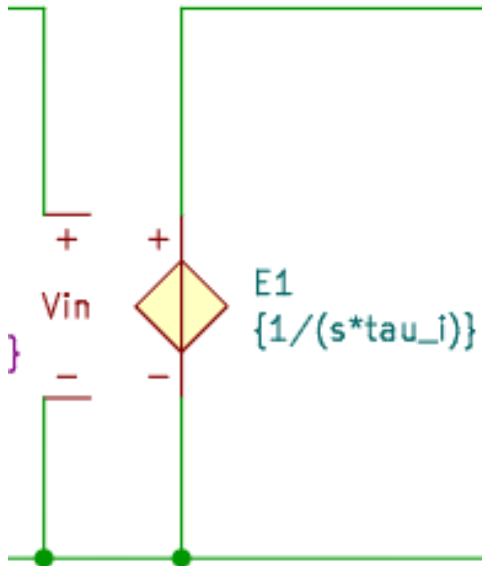
$$Q = 2\pi \cdot f_{res} \cdot C_s \cdot R_t$$

$$\text{Critically damped} \rightarrow Q = \frac{1}{\sqrt{2}}$$

$$R_t = \frac{Q}{2\pi \cdot f_{res} \cdot C_s} = 79.97 \cdot 10^3 \Omega$$

Ideal Amplifier Representation

Amplifier
concept



Sensitivity of pick up coil @1kHz is -59.4dBV/(A/m)

$$V_{ind_{1kHz}} = 10^{-59.4/20} = 1.07 \cdot 10^{-3} \text{ Vm/A}$$

Knowing from Laplace transform, $\frac{d}{dt} \leftrightarrow s = j\omega$, therefore

$$V_{ind} = k \cdot \frac{dH}{dt} \leftrightarrow V_{ind}(s) = k \cdot s \cdot H(s)$$

With that property, the conversion gain of the pick up coil can be written as:

$$V_{ind} = 1.07 \cdot 10^{-3} \frac{s}{2\pi \cdot 1000} = 0.171 \cdot 10^{-6} \cdot s \frac{\text{Vm}}{\text{As}}$$

Converting the units into Pa,

$$V_{ind} = \frac{0.171}{0.633} 10^{-6} s = 0.269 \cdot 10^{-6} \cdot s \frac{\text{V}}{\text{sPa}}$$

Sensitivity of the microphone @1kHz is -35.5dBV/Pa

$$V_{mic} = 10^{-35.5/20} = 0.0168 \frac{\text{V}}{\text{Pa}}$$

So, the gain A_{v1} of A1:

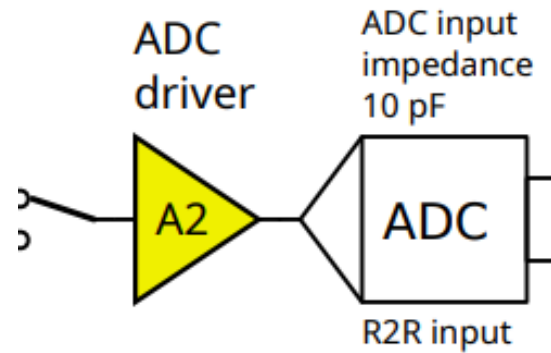
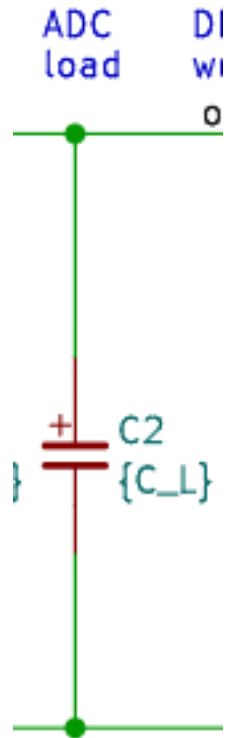
$$A_{v1} = \frac{V_{mic}}{V_{ind}} = \frac{0.0168}{0.269 \cdot 10^{-6}} = \frac{62.4 \cdot 10^3}{s}$$

Transfer function of E1

$$H(s) = \frac{1}{s\tau_i} = \frac{62.4 \cdot 10^3}{s}$$

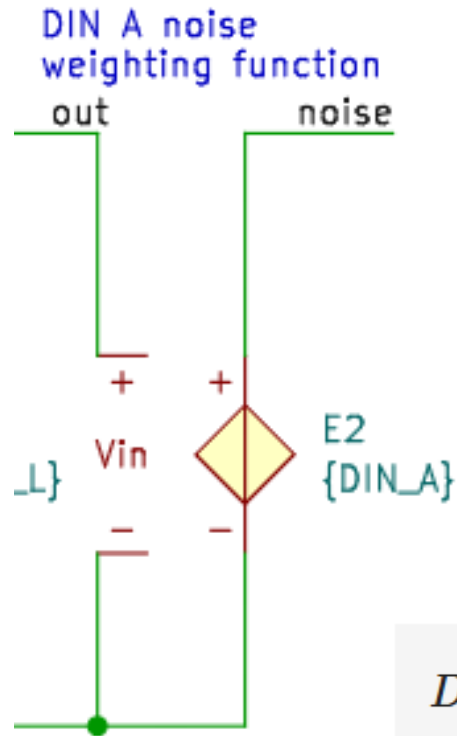
$$\tau_i = \frac{1}{62.4 \cdot 10^3} = 16.06 \cdot 10^{-6}$$

Output Representation



$$C_L = 10pF$$

Noise Representation



n_{SPL} in dBSPL converted into Pascals:

$$P_{noise} = 20 \times 10^{-6} \times 10^{\frac{n_{SPL}}{20}} Pa$$

Convert Pascals to Volts using microphone sensitivity

$$V_{onoise} = P_{noise} \times 10^{A_{mic}/20} = 20 \times 10^{-6} \times 10^{(A_{mic}+n_{SPL})/20}$$

With n_{SPL} of microphone is 30dBSPL and A_{mic} is -35.5 dBV/Pa

$$V_{onoise} = 10.62 \cdot 10^{-6} V$$

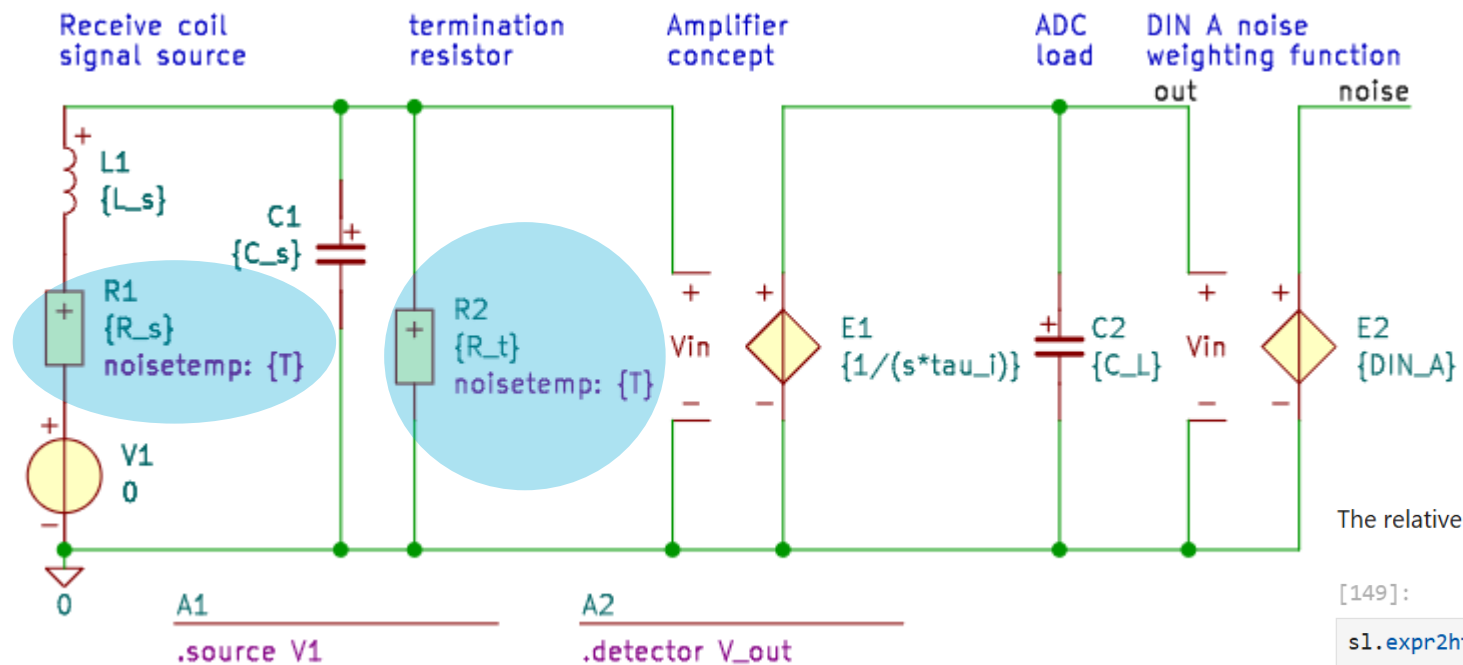
DIN_A

Noise weighting function

$$\frac{1.872 \cdot 10^8 f^4}{((f^2 + 1.16 \cdot 10^4)(f^2 + 5.445 \cdot 10^5))^{0.5} (f^2 + 424.4)(f^2 + 1.487 \cdot 10^8)}$$

Noise Budgeting

For initiation, deciding on **the noise budgeting of the signal source is crucial**



The relative part of the output noise taken by the source is:

[149]:

```
s1.expr2html(RMSnoise/V_onoise)
```

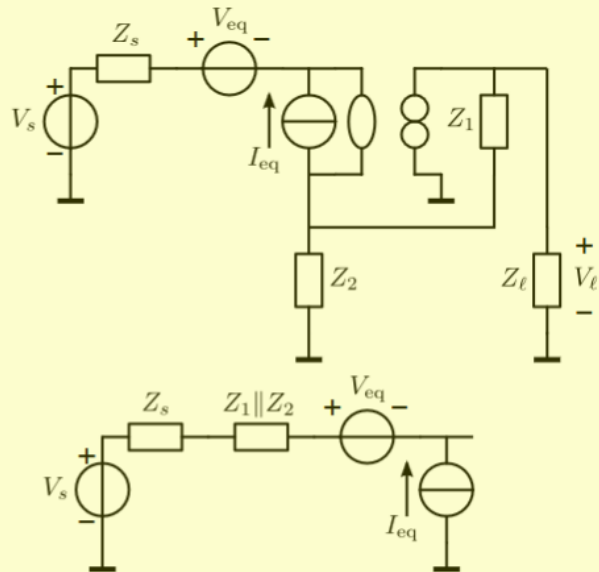
[149]:

0.1409

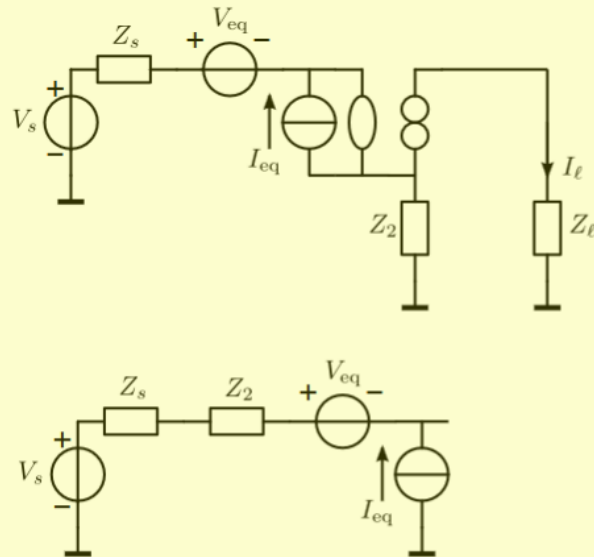
Feedback Design

Single-loop passive feedback configurations and their equivalent noise models

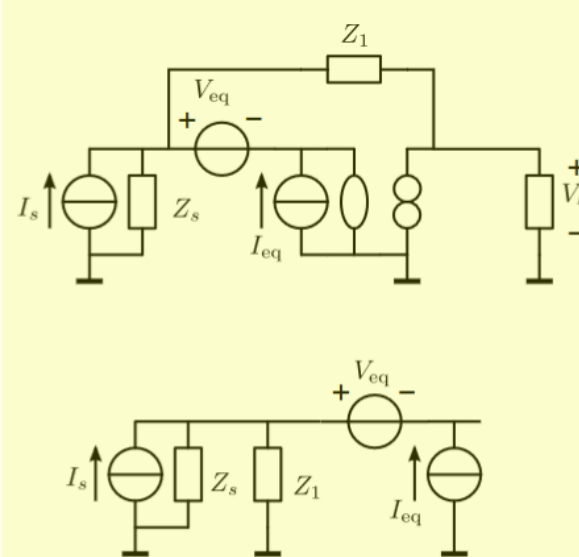
Voltage amplifier



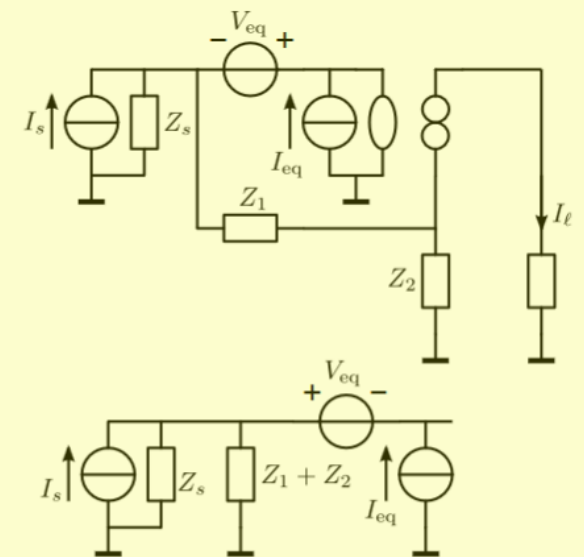
Transadmittance



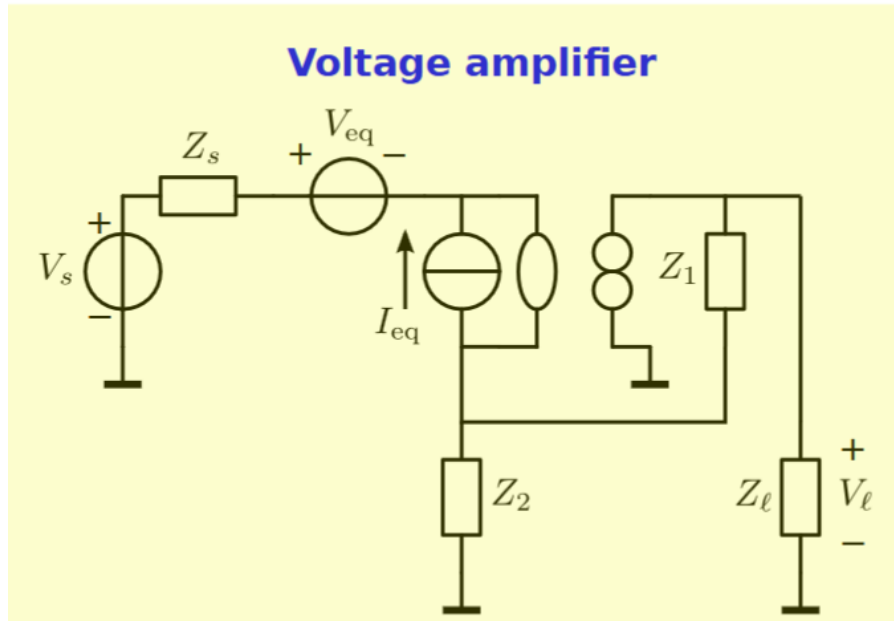
Transimpedance



Current amplifier



Feedback Design



- Selected configuration as voltage as input, and voltage as output
- Overall transfer an integrator with time constant τ_i
- Two choices for components of Z_1 and Z_2 respectively
- Option 1: Z_1 = Capacitor, Z_2 = Resistor
- Option 2: Z_2 = Inductor, Z_1 = Resistor
- Tradeoffs in terms of feasible inductor size, and corresponding resistor size (i.e more noise)
- Tradeoff in terms of manufacturability

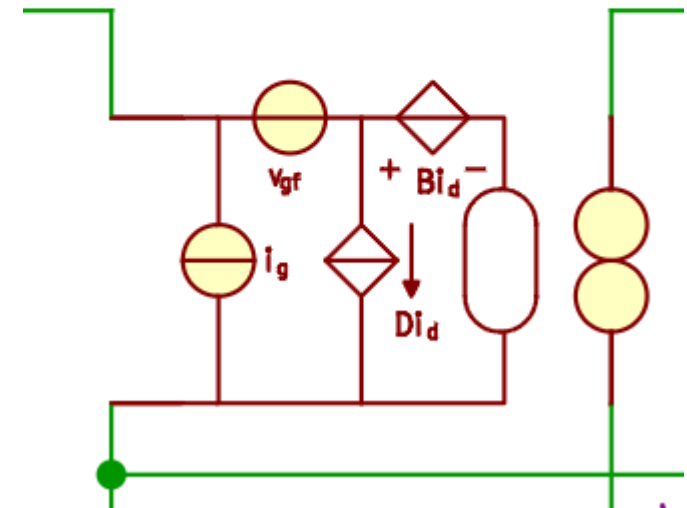
Feedback Design

- Noise budget of the feedback network limits the maximum value of the feedback resistor
- Power budget of the feedback network limits the minimum value of the feedback resistor
- A suitable value for the resistor component can be chosen, and then the capacitor value selected to match the specific integrator time constant defined by the specifications
- A bypass resistor is added in parallel with the capacitor
- This bypass resistor specifies the DC path of the circuit, and sets the lower corner frequency of the amplifier to $f_{\min}$

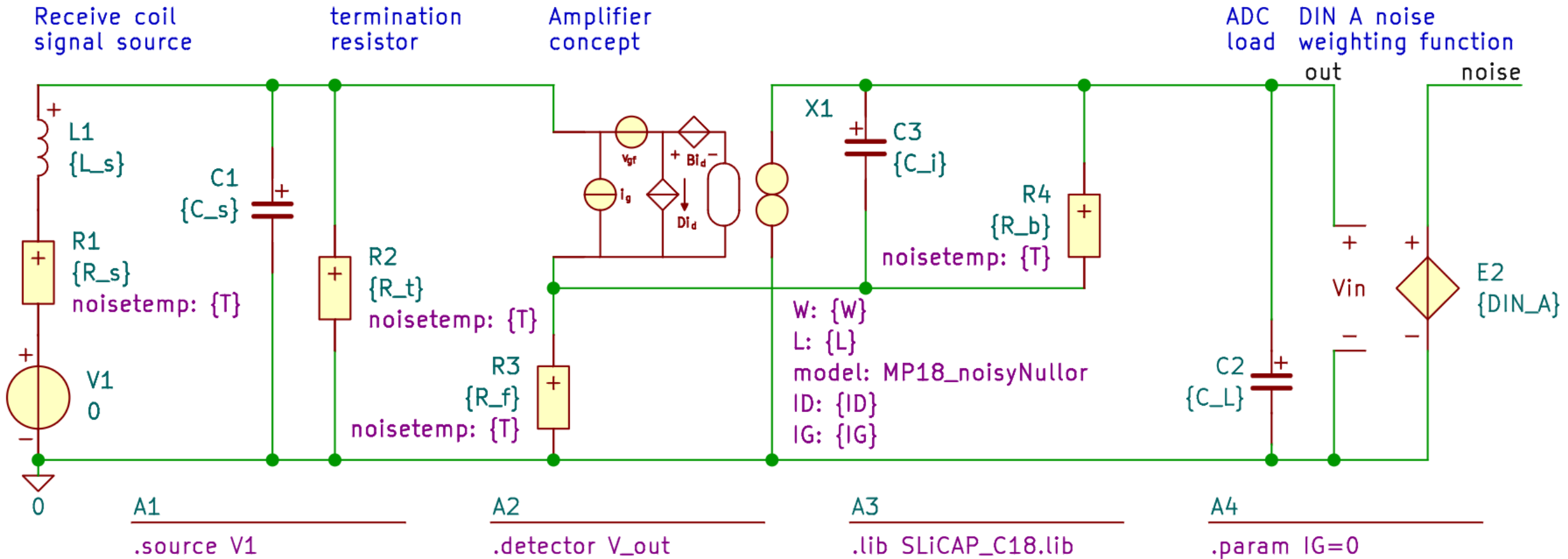
1st stage - Gm, Ciss, Inversion and Noise

- Why is noise so important? - I/p referred noise is set by the 1st stage
- Gm – Defines how 'strongly' the I/p stage is driven, also helps in thermal noise ($4kTy/gm$) (more current for less voltage is desirable)
- Ciss – More Ciss, more loading - If we increase device width for more Gm, Ciss increases :(
- Inversion – Weak inversion (large Gm/ I_d) would be preferred for maximum transconductance gain
- To implement the 1st stage, we need to model noise, so we replace nullor by noisy nullor

Amplifier
concept



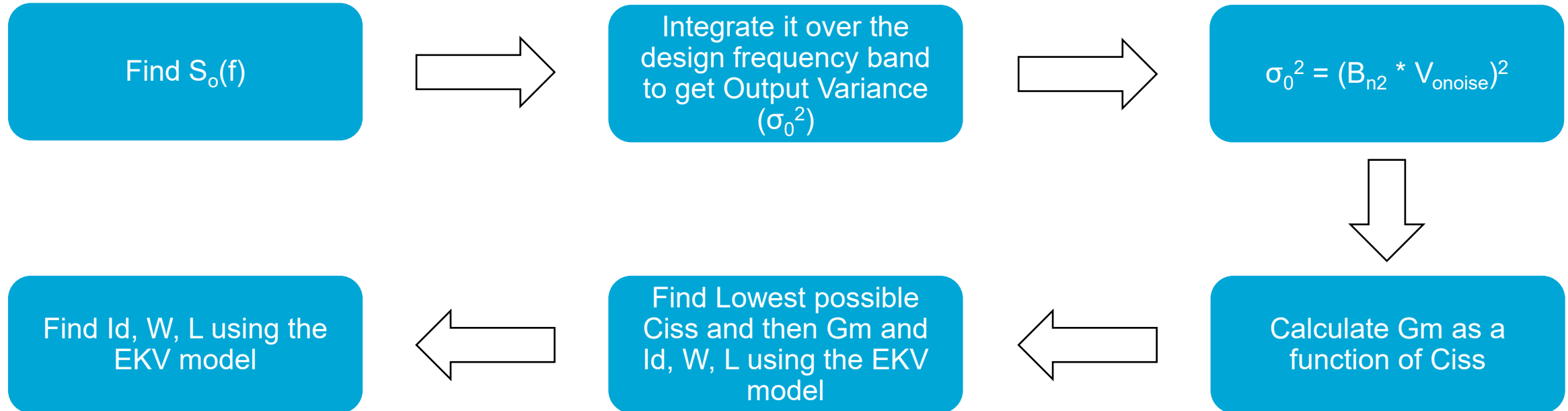
1st stage concept



What stage to implement the 1st stage as?

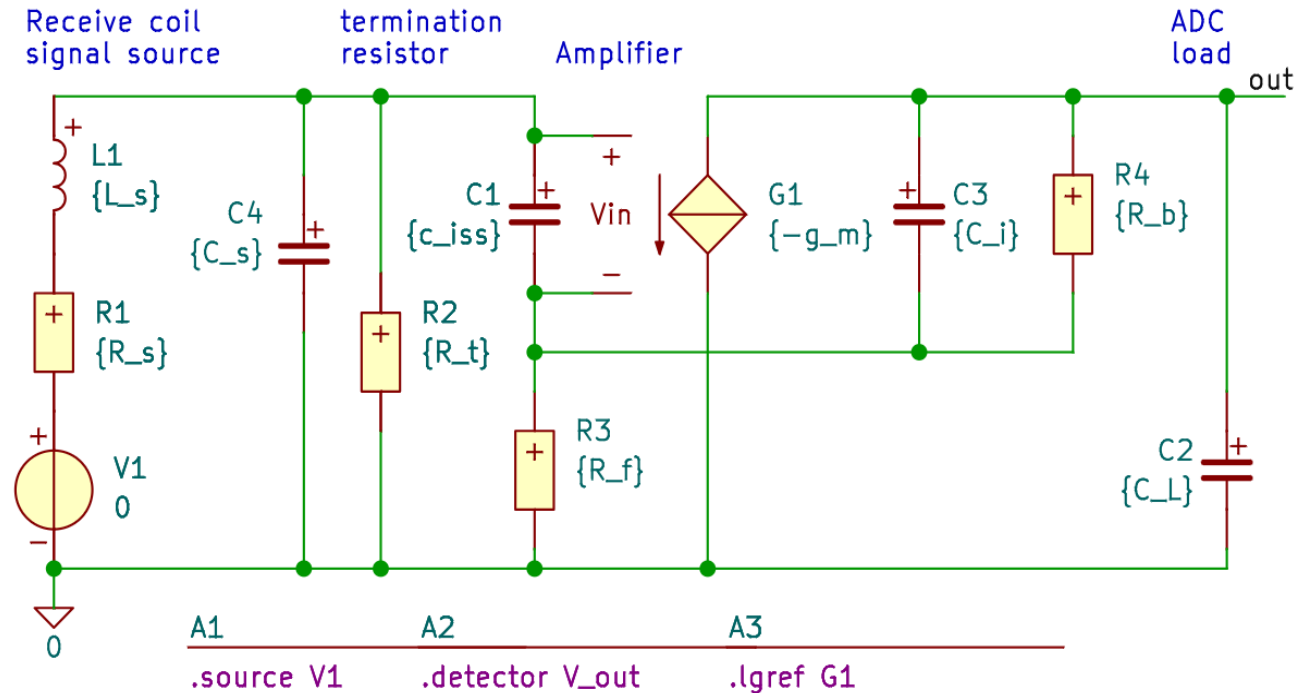
- Between CS, CG and CD we choose to design the first stage with a CS stage because of its T1 matrix.
- T1 matrix of a CS stage has the least values – showing its most nullor-like properties (noise performance of subsequent stages can be removed)
- Even if we can model CS, CG and CD to have the same noise, CD and CG are voltage and current followers, so we won't get any gain.
- SO, we choose to model the 1st stage with a CS stage amplifier (In saturation – max gain)

Budget to parameter calculations



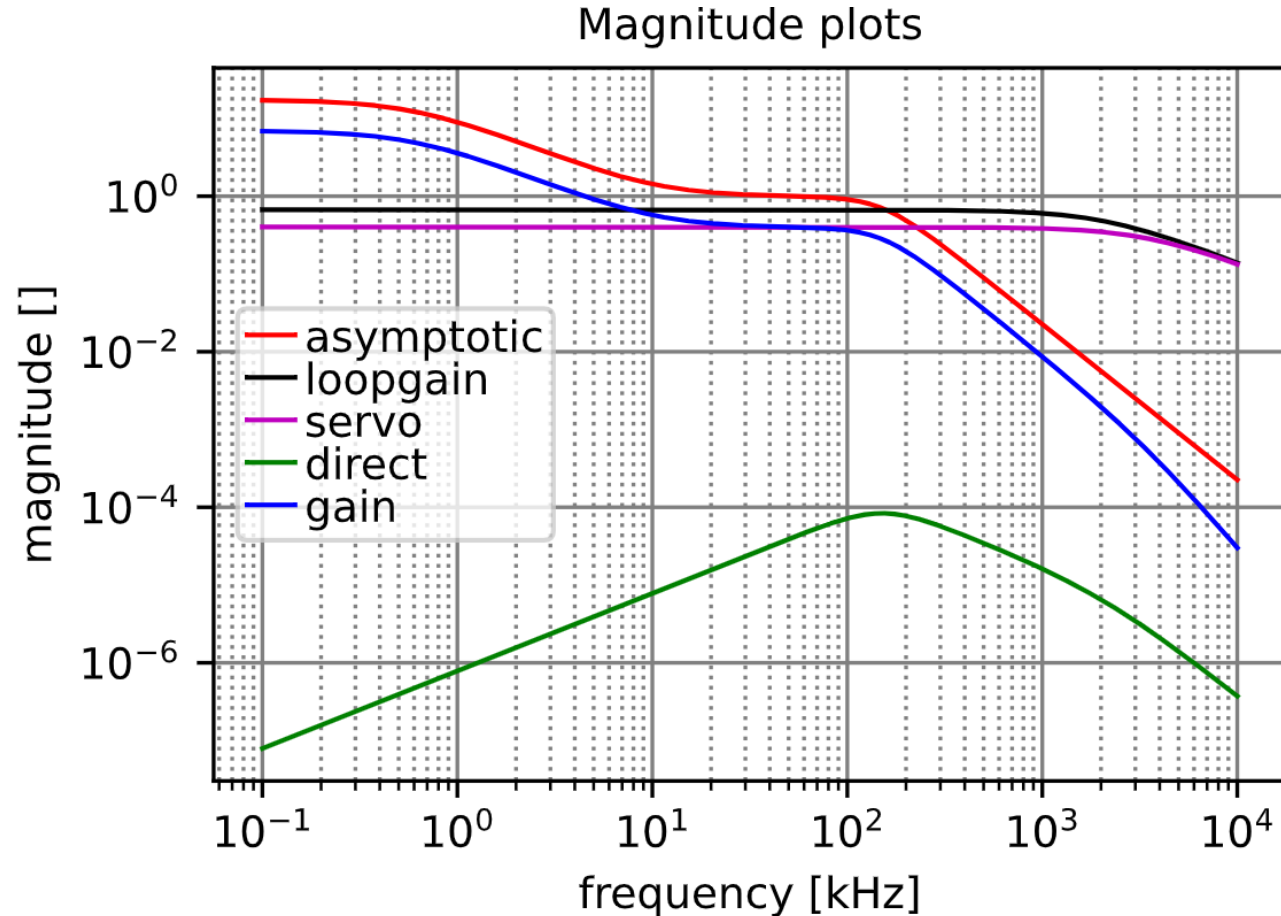
Single Stage

- From Part 3: g_m , c_{iss} , W , L , I_{DS}
- Check if one stage is enough
- Single CS = only 3 usable terminals
- → use a differential pair
- $\approx 4\times$ current, $4\times$ area vs. single CS
- Current drive check: ✓
- Voltage headroom check: ✓
- Loop gain analysis:
 - $|L_{DC}|$ below unity across all frequencies
 - Poor accuracy
 - → move to dual-stage design

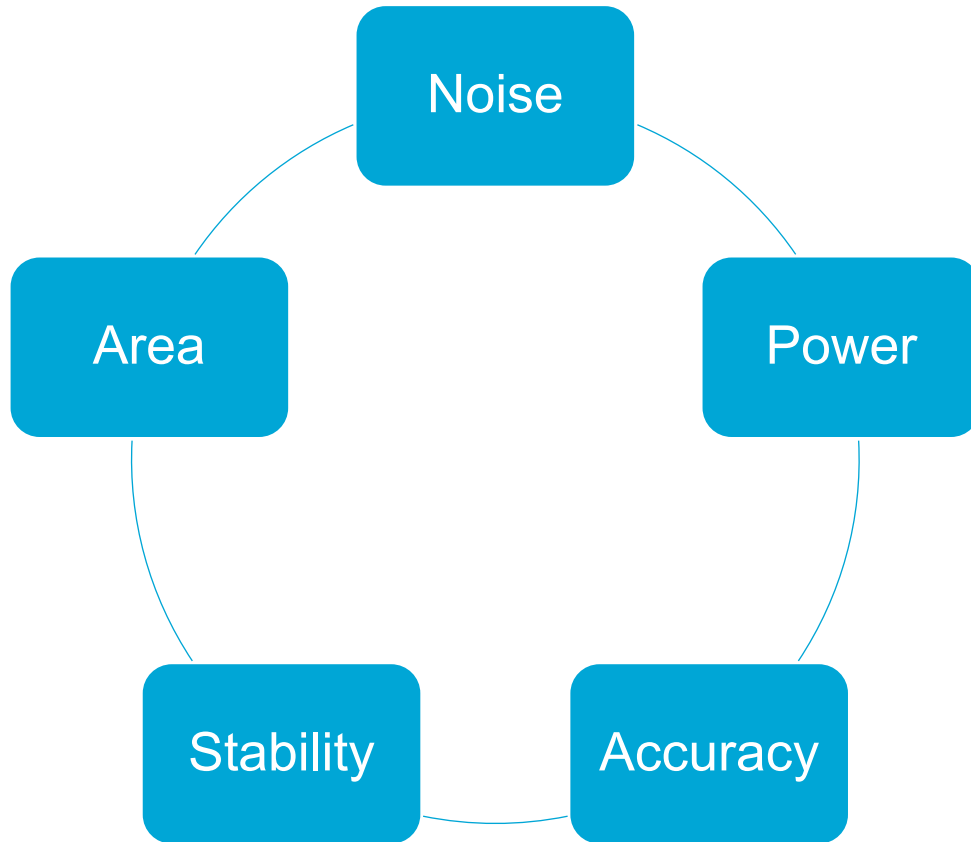


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Why Dual Stage?

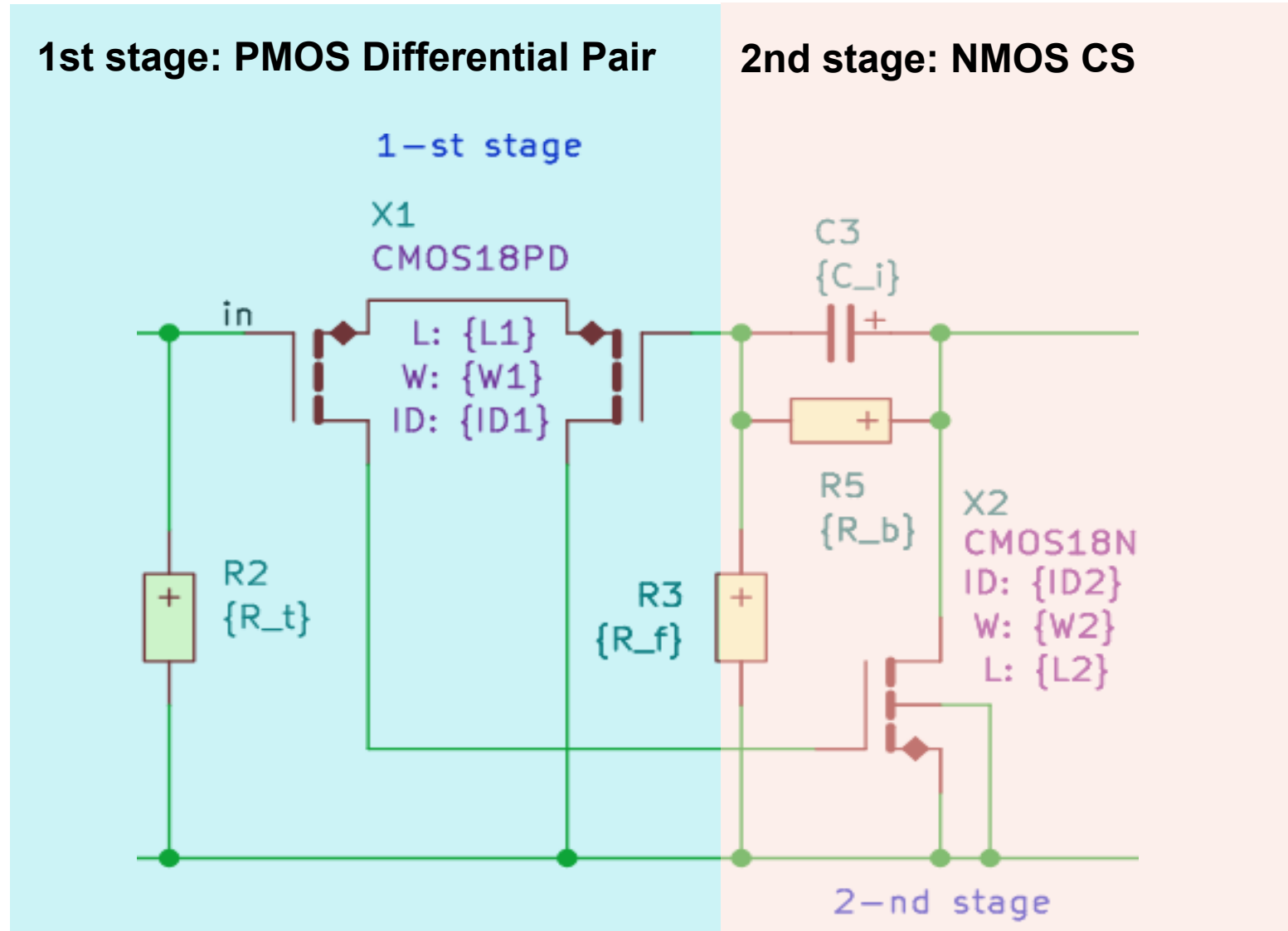


First Stage dominates noise

Second Stage dominates loop gain

Why Dual Stage?

Suggested Topology



First Stage

	Differential Pair	Single-Ended CS
✓	Lower Noise	Less Area
✓	High CMRR	Lower Power Consumption
✓	Matches Coil's 2 Terminal Input	Simple Design
—	More Area	Allows CM interference
—	Higher Complexity	Higher Noise

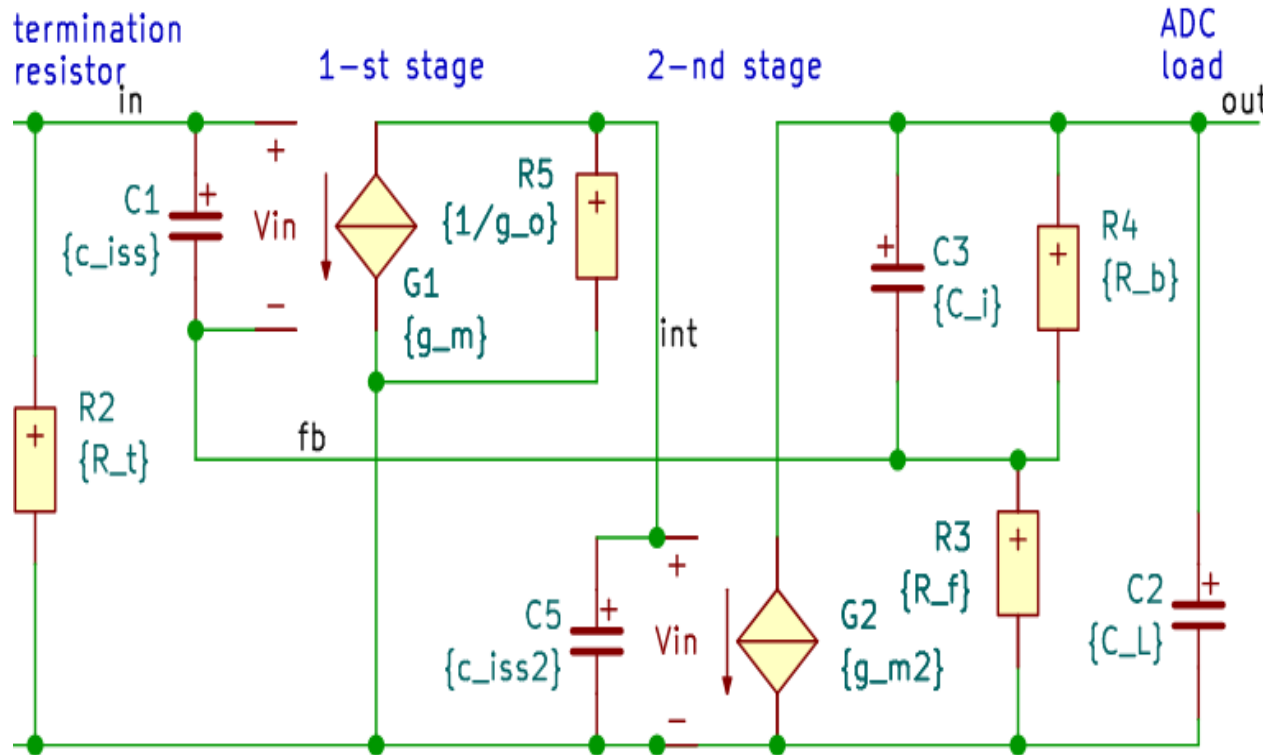
Second Stage

CHARACTERISTICS
High Loop Gain / Accuracy
High Current Drive Capability
Low Power Dissipation
Simple / Small Area



NMOS COMMON SOURCE

Dual Stage Design



$$L_{DC} = \frac{R_f g_{m1} g_{m2} (-R_s - R_t)}{R_s g_{o1} + R_t g_{o1}} = -\frac{R_f g_{m1} g_{m2}}{g_{o1}}$$

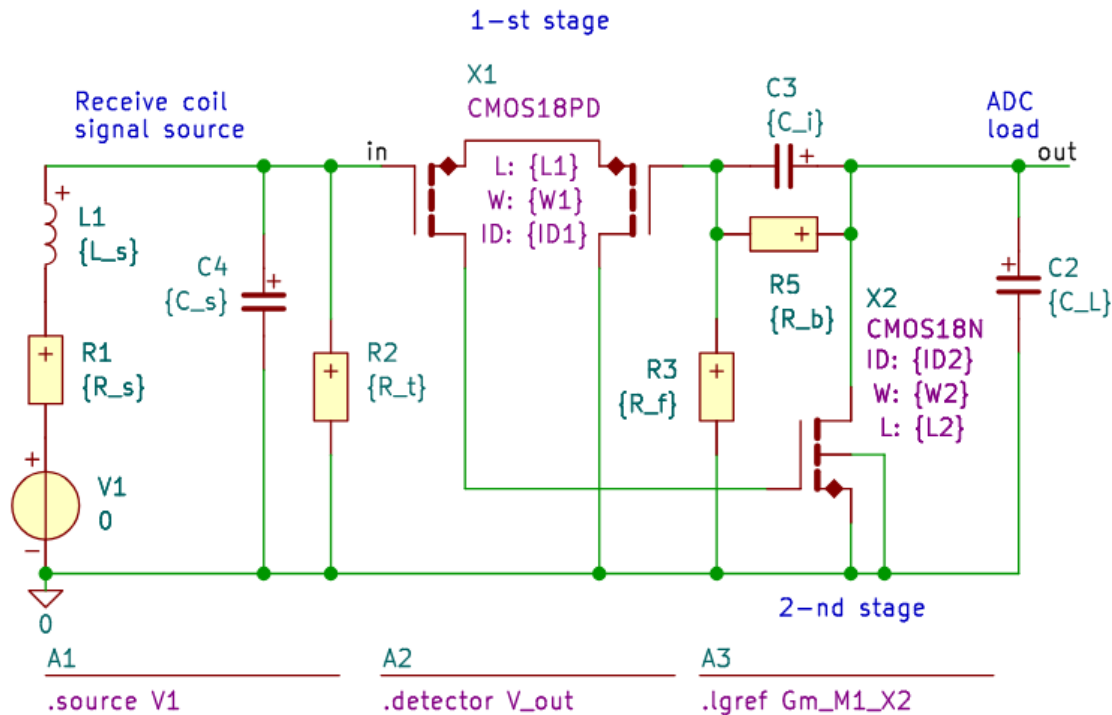
gm1 : First-stage Transconductance
Mainly set by the noise budget

gm2 : Second-stage Transconductance
Extra loop-gain factor

Rf : Feedback Resistor
Set by the desired transfer-function

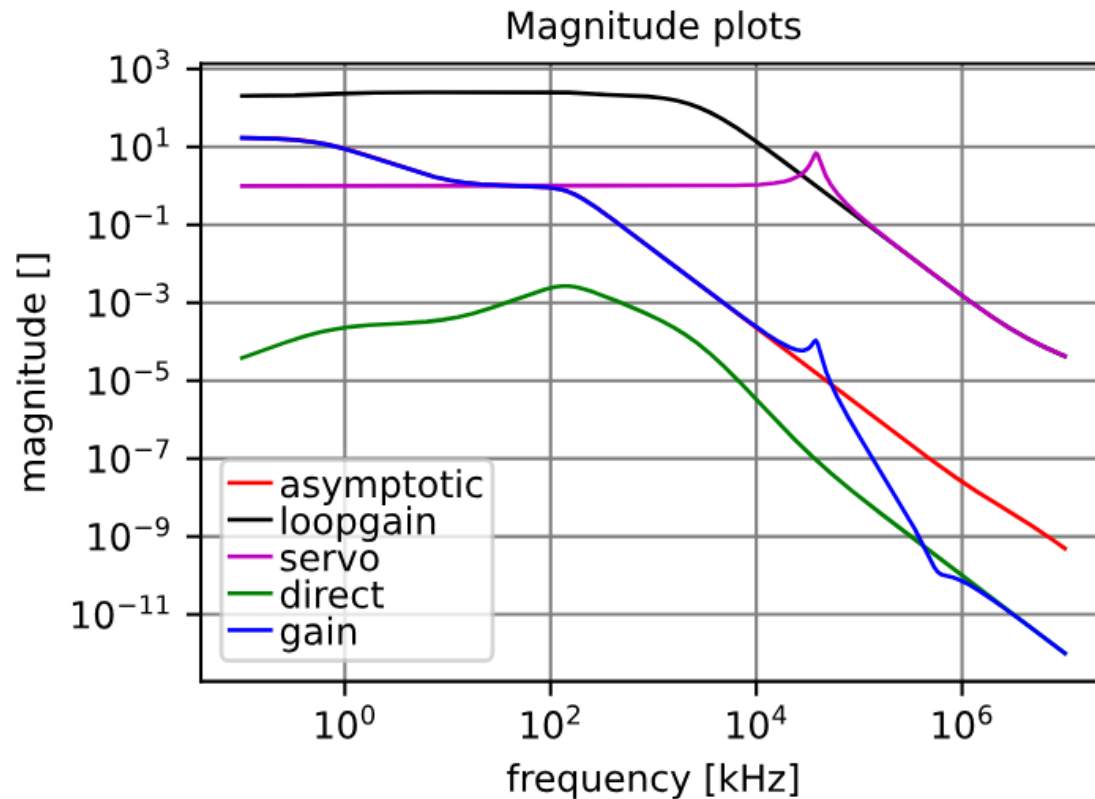
go1 : First-stage Output Conductance
Defines the first-stage output resistance

EKV Small Signal Model



- Realistic Transistor-level Design
- Includes most Parasitic Capacitances
- Results to more Accurate Gain Plots

Pole-Zero Analysis



pole	Re [Hz]	Im [Hz]	Mag [Hz]	Q
p ₁	-600.6		600.6	
p ₂	$-106.5 \cdot 10^3$	$106.6 \cdot 10^3$	$150.7 \cdot 10^3$	0.7076
p ₃	$-106.5 \cdot 10^3$	$-106.6 \cdot 10^3$	$150.7 \cdot 10^3$	0.7076
p ₄	$-2.756 \cdot 10^6$	$38.29 \cdot 10^6$	$38.39 \cdot 10^6$	6.966
p ₅	$-2.756 \cdot 10^6$	$-38.29 \cdot 10^6$	$38.39 \cdot 10^6$	6.966

zero	Re [Hz]	Im [Hz]	Mag [Hz]	Q
z ₁	$118.8 \cdot 10^6$	$559.2 \cdot 10^6$	$571.7 \cdot 10^6$	2.406
z ₂	$118.8 \cdot 10^6$	$-559.2 \cdot 10^6$	$571.7 \cdot 10^6$	2.406
z ₃	$-10.51 \cdot 10^3$		$10.51 \cdot 10^3$	

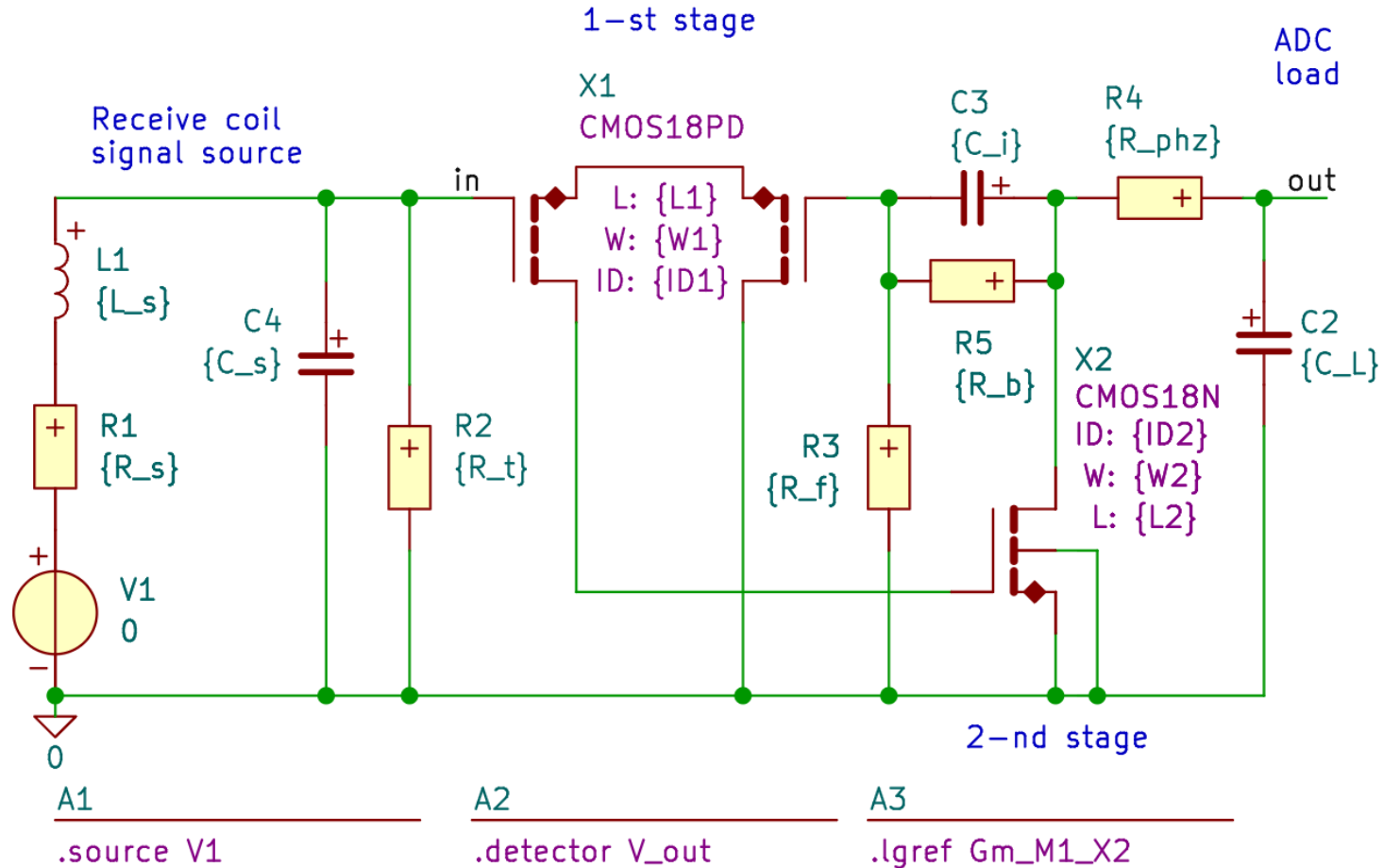
- The **Gain** plot follows the results of the poles and zeros as expected
- At around 40 MHz there is a peaking caused by the quality factor (Q) of P_{4,5} which is quite larger than 1, meaning the pole pair is severely underdamped, leading to visible peaking in the plot.
- The circuit is stable, $\text{Re}(p) < 0$

Frequency Compensation

- Maximally Flat Magnitude
- Phantom zero method
 - Benefits
 - Implementation

$$z = -\frac{(1 - L_{DC}) p_1 p_2}{\sqrt{2(1 - L_{DC}) p_1 p_2} + p_1 + p_2}$$

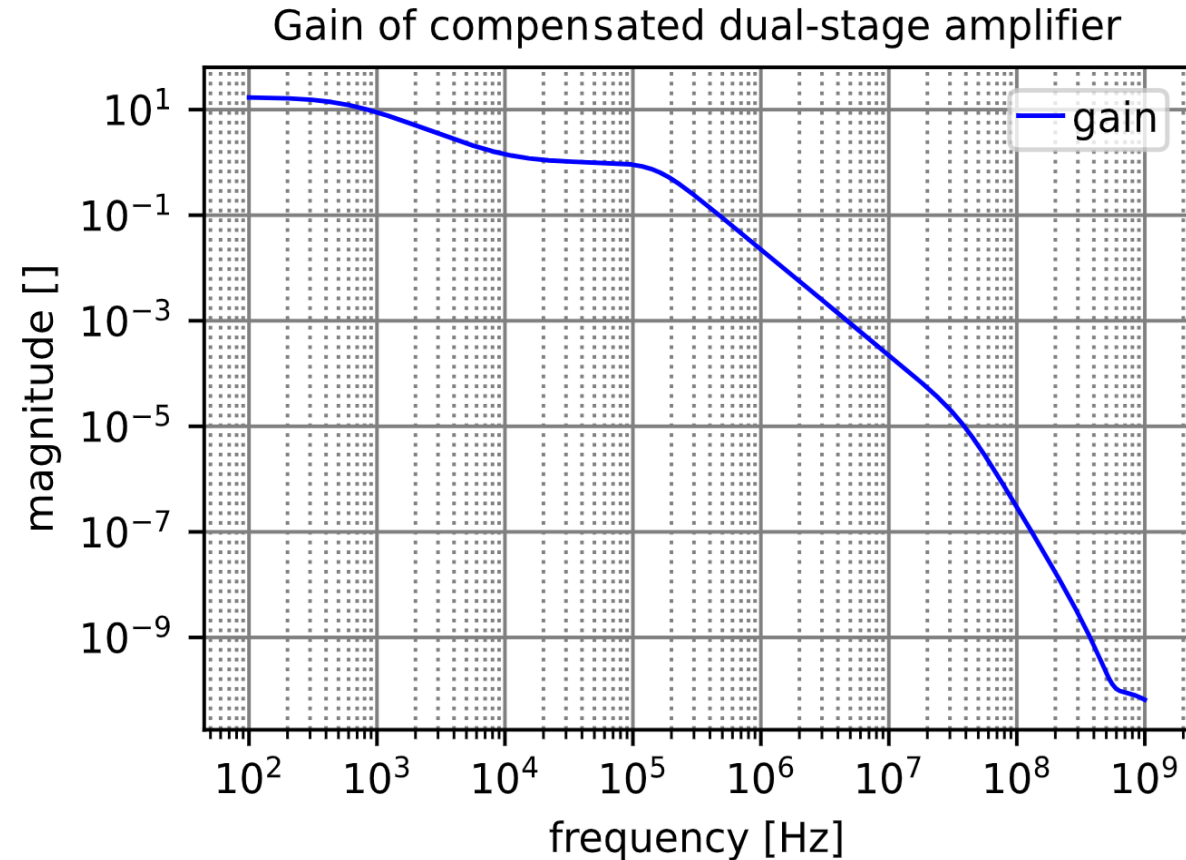
$$R_{phz} = -\frac{1}{zC_L}$$



Frequency Compensation

- Maximally Flat Magnitude

p_4	$-26.05 \cdot 10^6$	$26.38 \cdot 10^6$	$37.07 \cdot 10^6$	0.7116
p_5	$-26.05 \cdot 10^6$	$-26.38 \cdot 10^6$	$37.07 \cdot 10^6$	0.7116
p_6	$-5.662 \cdot 10^9$		$5.662 \cdot 10^9$	



Thank you

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